

41. The Principal Genus Theorem for Relative Galois Number Fields

Math. Ann. 108 (1933), pp. 411–419

In recent times it has become clear that noncommutative methods, in particular the theory of algebras, make it possible to formulate and prove familiar theorems on relative cyclic and abelian number fields for arbitrary relative Galois fields. I recall above all the norm theorem, which in general can be expressed as a theorem on split algebras. In what follows I show that, correspondingly, the principal genus theorem too can be formulated hypercomplexly, as a theorem on inner automorphisms or on crossed representations. The proof follows from the above-mentioned theorem on split algebras by purely algebraic-arithmetical considerations, whereas in the former theorem the norm theorem in the cyclic case, and already for prime degree, remains the transcendental core.

For better understanding of the formulation I first put forward the principal genus theorem “in the minimal sense”, not used below: it is an elementary algebraic theorem which, in the cyclic special case, passes into Hilbert’s familiar Theorem 90, according to which $N(a)$ is equal to one if and only if $a = b^{1-S}$. I formulate even this theorem as one about inner automorphisms or crossed representations, and prove it by means of the general theorems on inner automorphisms and crossed product representations of algebras.

In the actual principal genus theorem, the crossed product of the ideal-class group and the Galois group replaces the latter construction, but with a finer induced division of the factor systems into ideal classes. This captures an analogue of the ray-class division of class-field theory. Since general theorems on automorphisms and representations are still absent here, the principal genus theorem may be regarded as a first step in this direction. I therefore give a reduction, as mentioned above, by means of the theorem on split algebras. In this way the theorem is reduced to the corresponding theorem for ideals rather than ideal classes; the Brandt theorems on the interrelation of the maximal orders of an algebra may be regarded here as the analogue of the automorphism theorems. Together with the study of maximal orders of crossed products, they give a proof essentially parallel to the proof in the minimal case, though more complicated because one must pass to the individual places. I therefore prefer the almost trivial proof suggested by E. Artin, which is an even simpler analogue here of Speiser’s proof.

§1. Principal Genus Theorem in the Minimal Case

In this paragraph k denotes an arbitrary commutative field, not necessarily a number field; K/k is a separable Galois extension of degree n , and $\mathfrak{G}$ is its Galois group.

I first recall the familiar facts about crossed products. The crossed product means a simultaneous embedding of K and $\mathfrak{G}$ in an algebra A such that the automorphisms of K become inner. If $u_{S_1}, \dots, u_{S_n}$ are symbols corresponding to the n group elements, first set A , as a module of linear forms of rank n over K , equal to

$$A = u_{S_1}K + \dots + u_{S_n}K. \quad (1)$$

By demanding the inner automorphism generated by u_S , more generally by u_SK^* , A becomes a ring, hence an algebra of rank n^2 over k . The demand is expressed by

$$u_S^{-1}zu_S = z^S, \quad \text{or} \quad zu_S = u_Sz^S \quad (2)$$

for every $z \in K$, and by

$$u_Su_T = u_{ST}a_{S,T}, \quad a_{S,T} \in K^*. \quad (3)$$

The associative law is equivalent to

$$a_{S,T}^R a_{ST,R} = a_{S,TR} a_{T,R}. \quad (4)$$

Defining the product of two arbitrary elements by

$$\left(\sum_S u_S b_S\right) \left(\sum_T u_T c_T\right) = \sum_{S,T} u_{ST} a_{S,T} b_S^T c_T,$$

one obtains the crossed product of K with $\mathfrak{G}$ for the factor system $a_{S,T}$, written

$$A = (a_{S,T}, K, \mathfrak{G}).$$

One proves that A is a simple normal algebra over k , hence a matrix ring D_r of degree r over the associated division algebra D , and that K is a maximal commutative subfield, hence a splitting field. Conversely, for every such normal division algebra D there are matrix rings D_r which arise in this way as crossed products.

If one passes from u_S to $v_S = u_S c_S$, with $c_S \in K^*$, the associated factor systems are

$$\bar{a}_{S,T} = a_{S,T} \frac{c_S^T c_T}{c_{ST}}. \quad (5)$$

In particular the factor systems

$$\frac{c_S^T c_T}{c_{ST}}$$

are associated with one; these are called transformation quantities. Associated factor systems are collected into a class (a) . The classes (a) correspond one-to-one to direct product formations of an abelian group; under direct product they form an abelian group, isomorphic to the componentwise product of classes of factor systems. The identity element is the split algebra class, i.e. the system of all transformation quantities. This is the R. Brauer group of algebra classes.

For cyclic algebras one obtains the following special case. If Z/k is cyclic and S is a generating substitution, the powers of u correspond to the powers of S , and

$$A = Z + uZ + \cdots + u^{n-1}Z, \quad (1')$$

$$zu = uz^S, \quad u^n = \alpha, \quad (2'-3')$$

with

$$\alpha \in k^*, \quad \bar{\alpha} = \alpha N(c) \quad \text{if } v = uc. \quad (4'-5')$$

Thus each factor system here consists of a single element of the ground field, written $A = (\alpha, Z, S)$; the identity class is given by norms from Z^* , and the group of algebra classes is isomorphic to $k^*/N(Z^*)$.

The formulation of the principal genus theorem in the minimal case uses the fact that relations (2)–(5) are purely multiplicative and hence also define the extension group $\mathfrak{G}^*$ of $\mathfrak{G}$, consisting of the elements in the n complexes $u_S K^*$:

$$\mathfrak{G}^* = \{u_{S_1} K^*, \dots, u_{S_n} K^*\}. \quad (6)$$

Here K^* is an abelian normal divisor of $\mathfrak{G}^*$, and $\mathfrak{G}^*/K^* \simeq \mathfrak{G}$. The group $\mathfrak{G}^*$ consists of exactly the regular elements g^* which transform K into itself, i.e. $g^{*-1} K g^* = K$. Thus the ring automorphisms of K are produced by all and only the elements of $\mathfrak{G}^*$. Indeed, if $v = uw$ induces such an inner automorphism of K , then w commutes with every element of K , and so lies in K^* , since K is maximal commutative.

Principal genus theorem in the minimal case. First form: every group automorphism of $\mathfrak{G}^*$ extending the identity automorphism of K^* is inner and is produced by an element of K^* . Equivalently, if the transformation quantities

$$\frac{c_S^T c_T}{c_{ST}}$$

all have the value one, then the c_S are symbolic $(1 - S)$ -th powers:

$$c_S = b^{1-S} = \frac{b}{b^S} \quad (S \in \mathfrak{G}).$$

Third form: the group $\mathfrak{G}$ has only one crossed representation class of degree one in K^* belonging to factor system one.

A representation $u_S \mapsto C_S$ is called crossed with factor system $a_{S,T}$ if

$$C_S^T C_T = C_{ST} a_{S,T}$$

holds. Two representations belong to the same class if

$$C_S = B^{-S} D_S B.$$

I prove the first form and then show its equivalence to the second and third forms. The proof rests on the fact that every group automorphism of the indicated kind extends to a ring automorphism of A , and that

such an automorphism is known to be inner. Let v_S be the images of the u_S under this automorphism. Then $\sum v_S K$ again forms the crossed product of $\mathfrak{G}$ with K for the same factor system, because by assumption all relations (2)–(4) remain valid. The map

$$\sum u_S b_S \mapsto \sum v_S b_S$$

is a ring automorphism fixing all elements of K . Hence it is produced by an element of K^* .

The passage to the second form is based on the fact that, by (2), the v_S must have the form $v_S = u_S c_S$. Since the factor systems are also preserved, (5) says that the corresponding transformation quantities must all be one. Conversely, (2)–(5) show that the substitution $v_S = u_S c_S$ produces an automorphism of the required kind. The first form therefore yields

$$v_S = b^{-1} u_S b = u_S \frac{b}{b^S}, \quad c_S = b^{1-S}.$$

In the cyclic case the hypothesis says in particular $N(c) = 1$, and $c = b^{1-S}$ is the familiar theorem. The third form is immediately equivalent to the second: the hypothesis says that $u_S \mapsto c_S$ is a crossed representation of degree one with factor system one, and $c_S = b^{1-S}$ says that this representation is equivalent to the identity representation.

§2. The Principal Genus Theorem

Preliminary remarks on class divisions. In the principal genus theorem of class-field theory for cyclic relative fields, two different divisions into classes occur: absolute ideal classes in the upper field and ray classes in the lower field. In the general case this corresponds to the fact that a division of the ideals of the upper field into classes induces a finer division of the factor systems into ideal classes.

Let $\mathfrak{I}$ first denote the group of all ideals of K . Since $\mathfrak{G}$ induces automorphisms of $\mathfrak{I}$, the extension by $\mathfrak{G}$ is defined by relations (2)–(5), and consists of the n complexes $\{\cdots u_S \mathfrak{I} \cdots\}$. Passing from $\mathfrak{I}$ to the group of absolute ideal classes by imposing equivalence, the n complexes $u_S \tilde{\mathfrak{I}}$ are still defined, because $\mathfrak{G}$ also induces automorphisms of the ideal-class group, the principal class being transformed into itself. Thus the equivalence relation from $\mathfrak{I}$ to $\tilde{\mathfrak{I}}$ transfers from $\{\cdots u_S \mathfrak{I} \cdots\}$ to $\{\cdots u_S \tilde{\mathfrak{I}} \cdots\}$. In particular u_S is equivalent to all products $u_S(c_S)$, where (c_S) runs through principal ideals.

If now one requires the product relations (3) to be unambiguous in this equivalence sense, this says that all transformation quantities arising from principal ideals become uniquely equivalent. In other words:

In the system of the n complexes $\{\cdots u_S \tilde{\mathfrak{I}} \cdots\}$, where $\tilde{\mathfrak{I}}$ is the group of absolute ideal classes of K and H denotes the principal class, the product relations (3) for the $u_S H$ are unambiguously defined if and only if, in the induced ideal-class division of the factor systems, the principal class contains all transformation quantities arising from principal ideals.

Definition. The induced ideal-class division of the factor systems is fixed by requiring the principal class to consist of all principal ideals $(a_{S,T})$ for which, for at least one system of basis elements $a_{S,T}$, the algebra

$$(a_{S,T}, K, \mathfrak{G})$$

splits at all ramified places of K .

These principal ideals form a group under the product of factor systems; the group contains the transformation quantities

$$\frac{(c_S)^T (c_T)}{(c_{ST})},$$

because the factor systems $c_S^T c_T / c_{ST}$ produce algebras which split everywhere.

Formulation of the principal genus theorem. I state the theorem, corresponding to the minimal theorem, in three equivalent forms.

First form: with the induced ideal-class division of the factor systems taken as basis, suppose that the substitution

$$v_S = u_S c_S$$

produces an automorphism of the n complexes $\{\cdots u_S \tilde{\mathfrak{I}} \cdots\}$, i.e. that the same relations (3) hold for the v_S in the sense of this class division. Then this automorphism is inner and is produced by an ideal class b .

Second form: if the transformation quantities formed from the ideal classes c_S ,

$$\frac{c_S^T c_T}{c_{ST}},$$

all belong to the principal class of the induced class division of the factor systems, then the vectors $\{\cdots c_S \cdots\}$ of ideal classes form the principal genus, and the classes c_S are symbolic $(1 - S)$ -th powers: there is an ideal class b such that

$$c_S = b^{1-S} = \frac{b}{b^S} \quad (S \in \mathfrak{G}).$$

Third form: with the induced ideal-class division for the factor systems, the group $\mathfrak{G}$ has only one crossed representation class of degree one in the ideal-class group which belongs to the identity class of factor systems.

The equivalence of the three forms follows as in §1. The passage from the first to the second form is even simpler here, because the automorphism is already assumed to be produced by $v_S = u_S c_S$. For the third form one only has to note that representations in the ideal-class group are defined purely multiplicatively; this imposes no restriction for representations of degree one.

Proof of the principal genus theorem. The proof of the second form rests on two lemmas.

Lemma 1. If the transformation quantities formed from the ideals c_S ,

$$\frac{c_S^T c_T}{c_{ST}},$$

give the unit ideal, then the c_S are symbolic $(1 - S)$ -th powers:

$$c_S = b^{1-S} \quad (S \in \mathfrak{G}).$$

This is equivalent to the first form when the automorphism there is assumed to be produced by $v_S = u_S c_S$. Proof: $b = \sum c_S$ has the required property; the sum is the module sum, i.e. the greatest common divisor. Indeed,

$$b = \sum_S c_S, \quad b^T = \sum_S c_S^T, \quad b^T c_T = \sum_S c_S^T c_T = \sum_S c_{ST} = b,$$

and therefore $c_T = b^{1-T}$.

Lemma 2. Let the algebra

$$A = (a_{S,T}, K, \mathfrak{G})$$

split at all finite and infinite ramified places of K , and suppose that the principal ideals $(a_{S,T})$ are transformation quantities of ideals:

$$(a_{S,T}) = \frac{c_S^T c_T}{c_{ST}}.$$

Then A splits everywhere, absolutely. Thus the $(a_{S,T})$ are transformation quantities of principal ideals:

$$(a_{S,T}) = \frac{(d_S)^T (d_T)}{(d_{ST})}.$$

Conversely, every algebra which splits everywhere satisfies this condition.

The proof rests on the known behavior of A under p -adic extension. Let k_p be the p -adic extension of k , and $K_{\mathfrak{P}}$ the $\mathfrak{P}$ -adic extension of K , where $\mathfrak{P}$ is a prime ideal above p . Let A_p be the extension of A obtained by passing from k to k_p . Let $\mathfrak{Z}$ be the decomposition group of $\mathfrak{P}$, and $a_{\mathfrak{Z}}$ the corresponding part of the factor system. Then

$$A_p \sim (a_{\mathfrak{Z}}, K_{\mathfrak{P}}/k_p, \mathfrak{Z}).$$

If p is unramified, so that $\mathfrak{Z}$ is cyclic, this is the distinguished representation of A_p by means of the unramified inertia field $K_{\mathfrak{P}}/k_p$.

It remains to show that, under the hypothesis of Lemma 2, A_p also splits in this unramified case. For infinite places $K_{\mathfrak{P}} = k_p$, and A_p therefore splits. For finite p it follows first that the factor system $a_{\mathfrak{Z}}$

is associated with a factor system consisting of units. In passing to the normalized cyclic representation $A_p = (\alpha, K_{\mathfrak{P}}/k_p, R)$, with R a generator of $\mathfrak{Z}$, the factor system is expressed by the relations

$$u_{\mathfrak{P}_i} u_{\mathfrak{P}_j} = u_{\mathfrak{P}_{i+j}} \varepsilon_{\mathfrak{P}_i, \mathfrak{P}_j}.$$

Putting $u_R = u$, one obtains

$$u^i = u_R^i \varepsilon_R \varepsilon_{R^2} \cdots \varepsilon_{R^{i-1}},$$

and in particular $u_E = \varepsilon_E$, where E is the order of $\mathfrak{Z}$. Since in the unramified extension $K_{\mathfrak{P}}/k_p$ all units are norms, A_p splits. Hence A splits everywhere, and thus, by the theorem on split algebras, splits absolutely.

Equivalently, the hypothesis implies that $a_{S,T}$ are transformation quantities of elements,

$$a_{S,T} = \frac{d_S^T d_T}{d_{ST}}, \quad d_S \in K^*.$$

In weakened form this says that the $(a_{S,T})$ are transformation quantities of principal ideals. Conversely, if A splits everywhere, then the principal ideals $(a_{S,T})$ are transformation quantities of ideals at every place; this is just another formulation of the hypothesis in the theorem on split algebras.

The two lemmas now give the proof of the principal genus theorem. Let c_S be an ideal in the class c_S . The hypothesis says, by the definition of the induced class division, that the transformation quantities of the c_S become principal ideals $(a_{S,T})$ satisfying the hypotheses of Lemma 2. Hence there are principal ideals (d_S) such that the transformed quantities

$$\frac{c_S^T c_T}{c_{ST}}$$

become the unit ideal. The ideals $c_S/(d_S)$, which lie in the same classes c_S , therefore satisfy the hypothesis of Lemma 1. Thus an ideal class b exists with

$$c_S = b^{1-S}.$$

This means that the classes c_S are symbolic $(1 - S)$ -th powers of the class b .

In the cyclic case the principal genus theorem passes into the familiar theorem as soon as, after normalization, the principal class of the induced class division becomes the group of principal ideals (α) for which α is a norm residue at all ramified places.

(Received 27 October 1932.)

42. Split Crossed Products and Their Maximal Orders

Actualités scientifiques et industrielles 148 (1934), pp. 5–15

In the simplest case of split crossed products, namely those which yield split algebras and hence have factor systems associated with one, the relations can be followed rather explicitly. This remains true even for non-Galois splitting fields, i.e. maximal commutative subfields. The simple algebraic facts belonging here are developed in §1. In §2, with an algebraic number field as ground field, I give explicit representations of all maximal orders and of their ideals by means of modules and complementary modules of the underlying splitting field k . I further give a division into “domains”, in which all maximal orders are put into the same domain when they have the same intersection with k . Thus the domains correspond one-to-one to the orders in k ; the principal order is assigned the principal domain. The maximal orders of a domain are transformed into one another by ideals formed from the modules of the corresponding order; in the principal domain these are the extensions of ideals of k . In §3 I pass to arbitrary simple algebras. By refining the division into domains, requiring not only the same intersection but also agreement of the p -adic components of the maximal orders at ramified places, one can transfer the theorems obtained in the split case. In particular, the maximal orders of a principal domain are then also transformed into one another by extensions of ideals of k .

The theorems on the principal domain are found in Chevalley and Hasse; Chevalley also shows that without the refined division concerning the ramified places the theorems no longer hold in general.

§I. Matrix Units of Split Crossed Products and Complementary Bases of the Splitting Fields

In this paragraph Ω denotes an arbitrary ground field. Let k/Ω be a Galois separable extension of degree n , and let $\mathfrak{G}$ be the Galois group with elements $S, T, \dots$; let $u_S, u_T, \dots$ be the corresponding operators. We consider crossed products with factor system one, hence split algebras

$$K = \mathfrak{G} \times k = u_{S_1}k + \dots + u_{S_n}k, \quad u_S u_T = u_{ST},$$

with

$$zu_S = u_S z^S \quad (z \in k).$$

Put

$$E_{\mathfrak{G}} = \sum_{S \in \mathfrak{G}} u_S.$$

Then factor system one gives

$$E_{\mathfrak{G}} u_S = u_S E_{\mathfrak{G}} = E_{\mathfrak{G}}, \quad (1)$$

$$z E_{\mathfrak{G}} = \sum_S u_S z^S, \quad (2a)$$

and

$$E_{\mathfrak{G}} z E_{\mathfrak{G}} = E_{\mathfrak{G}} \sum_S u_S z^S = E_{\mathfrak{G}} \operatorname{Sp}(z), \quad (2b)$$

where $\operatorname{Sp}(z)$ is the trace of z as an element of k/Ω .

Theorem. If $a_1, \dots, a_n$ is a basis of k/Ω , and $\bar{a}_1, \dots, \bar{a}_n$ is the complementary basis, then

$$c_{ik} = \bar{a}_i E_{\mathfrak{G}} a_k$$

is a system of matrix units of K . Thus K can be represented as the module product

$$K = k E_{\mathfrak{G}} k = k \left(\sum_S u_S \right) k = \sum_{i,k} (\bar{a}_i E_{\mathfrak{G}} a_k) \Omega.$$

Indeed, by (2b),

$$c_{ij} c_{hk} = \bar{a}_i E_{\mathfrak{G}} a_j \bar{a}_h E_{\mathfrak{G}} a_k = \bar{a}_i E_{\mathfrak{G}} a_k \operatorname{Sp}(a_j \bar{a}_h) = c_{ik} \operatorname{Sp}(a_j \bar{a}_h),$$

which is precisely the matrix-unit property, since the complementary bases are defined by

$$\operatorname{Sp}(a_j \bar{a}_h) = \delta_{jh}.$$

Remark 1. If Z is any commutative extension field of Ω , everything remains valid when a_i is a basis of k_Z/Z and K_Z becomes a direct sum of fields. The substitutions of the group are then defined by their inducing the identity on Z .

Remark 2. That $K = k E_{\mathfrak{G}} k$ also follows directly from the fact that the right-hand side is a nonzero ideal of the simple system K .

Now assume that the ground field Ω has characteristic zero. The representation $K = k E_{\mathfrak{G}} k$ of the split algebra remains valid even when k is a non-Galois splitting field of K/Ω . One passes to the Galois splitting field $\bar{k}$. Let $\bar{k}$ be the Galois field belonging to k , $\bar{\mathfrak{G}}$ its group, $\bar{u}_S$ the corresponding operators, and $\bar{K} = \bar{k} E_{\bar{\mathfrak{G}}} \bar{k}$ a split crossed product. Let the subfield k correspond to the subgroup $\mathfrak{H}$ of order h . Set

$$\bar{\mathfrak{G}} = \mathfrak{H} S_1 + \dots + \mathfrak{H} S_n = S_1^{-1} \mathfrak{H} + \dots + S_n^{-1} \mathfrak{H}, \quad E_{\mathfrak{H}} = \frac{1}{h} \sum_{H \in \mathfrak{H}} u_H,$$

and

$$E_{\mathfrak{G}} = \frac{1}{h} E_{\bar{\mathfrak{G}}} = \sum_i u_{S_i}, \quad u_{S_i} = E_{\mathfrak{H}} \bar{u}_{S_i}.$$

Then $E_{\mathfrak{H}}$ generates the identity representation of $\mathfrak{H}$, while $E_{\mathfrak{G}}$, like $E_{\bar{\mathfrak{G}}}$, generates the identity representation of $\bar{\mathfrak{G}}$. For these $E_{\mathfrak{G}}$ and u_{S_i} , relation (1) holds on one side and (2) holds completely. First,

$$E_{\mathfrak{G}} = E_{\mathfrak{G}}E_{\mathfrak{H}} = E_{\mathfrak{H}}E_{\mathfrak{G}} = E_{\mathfrak{H}}E_{\mathfrak{G}}E_{\mathfrak{H}}, \quad (3)$$

$$zE_{\mathfrak{H}} = E_{\mathfrak{H}}z, \quad (4)$$

and hence

$$zu_{S_i} = u_{S_i}z^{S_i}. \quad (5)$$

Equation (2a) follows from (5) by summation, and then (2b) follows, while (1) holds on one side:

$$E_{\mathfrak{G}}u_{S_i} = E_{\mathfrak{G}}.$$

The proof of $K = kE_{\mathfrak{G}}k$ is now exactly the same as before. As a full matrix ring over Ω , K contains every field of degree n as a subfield, in particular k .

For §3 I also note that the passage from K to $\bar{K}$ is possible even when K does not split. Besides k , the field $\bar{k}$ is also a splitting field, and K is therefore similar to $\bar{K}$, the crossed product of $\bar{\mathfrak{G}}$ with $\bar{k}$. Since k is a splitting field, the subgroup $\mathfrak{H}$ corresponding to k has factor system one, so the idempotent $E_{\mathfrak{H}}$ is as above. It follows conversely that K is isomorphic to $E_{\mathfrak{H}}\bar{K}E_{\mathfrak{H}}$. This algebra is isomorphic to the automorphism ring of the left ideal $\bar{K}E_{\mathfrak{H}}$, and hence is similar to $\bar{K}$; the rank agrees with that of K . In the split case this gives again

$$E_{\mathfrak{H}}\bar{K}E_{\mathfrak{H}} = E_{\mathfrak{H}}\bar{k}E_{\bar{\mathfrak{G}}}\bar{k}E_{\mathfrak{H}} = (E_{\mathfrak{H}}\bar{k}E_{\mathfrak{H}})E_{\bar{\mathfrak{G}}}(E_{\mathfrak{H}}\bar{k}E_{\mathfrak{H}}) = kE_{\mathfrak{G}}k.$$

§II. The Maximal Orders of Split Crossed Products and Their Division into Domains

From now on Ω is a finite algebraic number field, K a split algebra of degree n over Ω , and k a Galois or non-Galois maximal commutative subfield, so that $K = kE_{\mathfrak{G}}k$. Let $a, \bar{a}$ be a pair of complementary finite $\mathfrak{o}$ -modules in k , where $\mathfrak{o}$ is the principal order of Ω and o the principal order of k . The structure of the maximal orders of K relative to k is described by the following theorems.

Theorem 1. The module product

$$\mathcal{O}_a = \bar{a}E_{\mathfrak{G}}a$$

is a maximal order of K ; in particular

$$\mathcal{O} = \mathcal{O}_o = oE_{\mathfrak{G}}o$$

is a maximal order.

Theorem 2. The transformation of $\mathcal{O}$ into $\mathcal{O}_a$ is effected by the left ideal

$$\mathcal{L} = oE_{\mathfrak{G}}a$$

and by its reciprocal right ideal

$$\mathcal{R} = \bar{a}E_{\mathfrak{G}}o$$

with respect to $\mathcal{O}$.

Remark. The occurrence of the two complementary modules $a, \bar{a}$ in the product $\mathcal{L}\mathcal{R}$ leads to the reciprocal relation precisely because of the trace formation caused by $E_{\mathfrak{G}}$.

Theorem 3. All left and right ideals with respect to $\mathcal{O}$ are exhausted by those described in Theorem 2. Consequently the orders $\mathcal{O}_a$ exhaust all maximal orders in K . Any two maximal orders $\mathcal{O}_a$ and $\mathcal{O}_b$ are transformed into one another by

$$\mathcal{L}_{ab} = \bar{a}E_{\mathfrak{G}}b, \quad \mathcal{R}_{ba} = \bar{b}E_{\mathfrak{G}}a,$$

and these exhaust the left and right ideals with respect to $\mathcal{O}_a$.

Theorem 4. As $\mathcal{O}_a$ runs through all maximal orders of K , the intersection

$$[\mathcal{O}_a, k]$$

runs through all orders of k ; this intersection is in each case the order of a . If all maximal orders belonging to the same order in k are grouped into one domain, then the transformations within a domain are given by left ideals $\bar{a}E_{\mathfrak{G}}b$ with respect to $\mathcal{O}_a$ and by reciprocal right ideals, where a and b are modules of the corresponding order in k .

Theorem 4a. In particular, the transformation of the maximal orders of the principal domain, i.e. the domain of all maximal orders containing the principal order o of k , is effected by ideals

$$aE_{\mathfrak{G}}b,$$

where a and b are ideals of k . Equivalently: if $\mathcal{L}$ is a left ideal of a maximal order of the principal domain and o is contained in the right order of $\mathcal{L}$, then $\mathcal{L}$ is the extension of an o -ideal.

The proofs rest on passage to the individual places. It is enough to pass to the quotient ring at p , where p runs through all prime ideals of Ω . If $\mathcal{C}$ is any $\mathfrak{o}$ -module in K , let its component $\mathcal{C}_p$ denote the extended module $\mathcal{C}\mathfrak{o}_p$. Then:

Lemma. $\mathcal{C}$ is the intersection of all extension modules $\mathcal{C}_p$. If $w \in \mathcal{C}_p$, there is a $\sigma \in \mathfrak{o}$, not divisible by p , such that $\sigma w \in \mathcal{C}$. Conversely, if w lies in the intersection of all $\mathcal{C}_p$ and g is the ideal of all $\sigma \in \mathfrak{o}$ with $\sigma w \in \mathcal{C}$, then g is divisible by no p , so $g = \mathfrak{o}$. No hypothesis on the rank of $\mathcal{C}$ is required.

Consequences. Every module is determined by its components. If $\mathcal{D}$ is a proper overmodule of $\mathcal{C}$, at least one $\mathcal{D}_p$ is a proper overmodule of $\mathcal{C}_p$. In particular, if $\mathcal{M}$ is an order in K , and all its components $\mathcal{M}_p$ are maximal orders over $\mathfrak{o}_p$, then $\mathcal{M}$ itself is maximal.

Proof of Theorem 1. $\mathcal{O}_a$ is an order: it is a finite $\mathfrak{o}$ -module of maximum rank. Moreover,

$$\mathcal{O}_a^2 = \bar{a}E_{\mathfrak{G}}a \bar{a}E_{\mathfrak{G}}a = \bar{a}E_{\mathfrak{G}}a \operatorname{Sp}(a\bar{a}) \subseteq \bar{a}E_{\mathfrak{G}}a = \mathcal{O}_a.$$

For maximality it suffices, by the consequence above, to prove maximality of every component. Since $\mathfrak{o}_p$ is a principal ideal ring, a_p and $\bar{a}_p$ have independent complementary bases $a_1, \dots, a_n$ and $\bar{a}_1, \dots, \bar{a}_n$. The products $\bar{a}_i E_{\mathfrak{G}} a_k$ form matrix units by §1; hence

$$\mathcal{O}_{a,p} = \sum_{i,k} (\bar{a}_i E_{\mathfrak{G}} a_k) \mathfrak{o}_p$$

is maximal. Since $\mathcal{O}_a$ is maximal, $\operatorname{Sp}(a\bar{a}) = \mathfrak{o}$ follows.

Proof of Theorem 2. From the trace rule one obtains

$$\begin{aligned} \mathcal{O}\mathcal{L} &= oE_{\mathfrak{G}}o oE_{\mathfrak{G}}a = oE_{\mathfrak{G}}a = \mathcal{L}, \\ \mathcal{R}\mathcal{O} &= \bar{a}E_{\mathfrak{G}}o oE_{\mathfrak{G}}o = \bar{a}E_{\mathfrak{G}}o = \mathcal{R}, \\ \mathcal{L}\mathcal{R} &= oE_{\mathfrak{G}}a \bar{a}E_{\mathfrak{G}}o = oE_{\mathfrak{G}}o = \mathcal{O}, \\ \mathcal{R}\mathcal{L} &= \bar{a}E_{\mathfrak{G}}o oE_{\mathfrak{G}}a = \bar{a}E_{\mathfrak{G}}a = \mathcal{O}_a. \end{aligned}$$

Proof of Theorem 3. Let $\mathcal{L}$ be a left ideal with respect to $\mathcal{O}$. Then

$$\mathcal{L} = oE_{\mathfrak{G}}\mathcal{L} = (oE_{\mathfrak{G}}l_1, \dots, oE_{\mathfrak{G}}l_n)$$

for an $\mathfrak{o}$ -basis $l_1, \dots, l_n$ of $\mathcal{L}$. Writing $l_\mu = \sum_i u_{S_i} z_{i\mu}$ gives $E_{\mathfrak{G}}l_\mu = E_{\mathfrak{G}}a_\mu$, and hence $\mathcal{L} = oE_{\mathfrak{G}}a$. If $\mathcal{L}$ is regular, then a has maximum rank n , and $\bar{a}E_{\mathfrak{G}}o$ is the reciprocal right ideal. Composition gives the general assertion.

Proof of Theorem 4. The intersection $t = [k, \mathcal{O}_a]$ is an order. As a subset of $\mathcal{O}_a$, it is a right multiplier domain and is the largest order in o with this property; it therefore contains the order o_a of a . To prove $t = o_a$, it suffices to prove equality componentwise. If $\tau \in t_p$, then τc_{ik} , for every $c_{ik} = \bar{a}_i E_{\mathfrak{G}} a_k$, is a linear combination of the c_{ij} with coefficients in $\mathfrak{o}_p$. On the other hand $a_k \tau$ is a linear combination of the a_j with coefficients in $\mathfrak{o}_p$. By the uniqueness of the representation through the c_{ij} , these coefficients must lie in $\mathfrak{o}_p$. Thus τ lies in the order of a_p . This proves that t is the order of a .

Proof of Theorem 4a. This is the specialization of Theorem 4 to the principal domain. Starting with $\mathcal{O} = oE_{\mathfrak{G}}o$, if o is contained in the right order of $\mathcal{L} = oE_{\mathfrak{G}}a$, then $\bar{a}E_{\mathfrak{G}}a$ belongs to the principal domain. Therefore a is an ideal of k , and $\mathcal{L} = \mathcal{O}a$. Starting from $\mathcal{O}_a$, correspondingly $\mathcal{L} = \bar{a}E_{\mathfrak{G}}b$, and $a^{-1}b$ is an ideal of k ; hence $\mathcal{L} = \mathcal{O}_a a^{-1}b$.

§III. Maximal Orders of Arbitrary Crossed Products

The theorems derived for split crossed products remain valid in the general case once the division into domains is refined. The facts about the principal domain then hold exactly; the remaining facts must be formulated place by place. Here a place is to be understood as a p -adic extension.

Definition. Let K be a simple normal algebra over Ω , and let k be a maximal commutative subfield. A domain relative to k consists of all maximal orders of K which have the same intersection with k and which, in addition, have the same p -adic components at the ramified places of K . If the intersection with k is the principal order, one obtains a principal domain.

For principal domains:

Theorem 1. The maximal orders of a principal domain are transformed into one another by transformation with extension ideals of ideals of k . Equivalently: the left ideals of the maximal orders of a principal domain which are prime to the different and contain o in their right order are extensions of ideals of k .

Theorem 2. If K has prime degree, there is only one principal domain; it is transformed into itself by extension ideals of the k -ideals. The corresponding ideals of a fixed maximal order $\mathcal{O}$ are the ideals of k extended by the two-sided ideals of $\mathcal{O}$.

Proof. Two maximal orders of a principal domain differ only at finitely many places; by definition these are places where K splits. There K has the form considered in §§1–2. This is clear when k is Galois; if the factor system is associated with one, it may be replaced by the system one. By §1 this also holds in the non-Galois case. Thus at every such place, and hence globally, the ideals are extension ideals.

If K has prime degree, it is either split or a division algebra. In the latter case it remains a division algebra at all ramified places and therefore has only one principal domain, so transformation by extension ideals again applies. The most general transformation is obtained by composing the extension ideals with those which transform the maximal order into itself, namely with two-sided ideals; these, as is known, differ from central ideals only at the ramified places, hence differ only by extension ideals. This proves the remaining assertions.

Also:

Theorem 3. The maximal orders of an arbitrary domain are transformed into one another by ideals prime to the different which, at every place, are composed from modules of the corresponding order as described in §2, Theorem 3. For a fixed ideal the components of these modules differ from the order at only finitely many places.

This follows directly from §2; one only has to replace the operators u_S at the individual places by corresponding equivalent operators.

Whether, for every order of k , there are maximal orders having that order as intersection remains to be investigated more closely at the ramified places. A principal domain always exists, as follows from the crossed, or possibly generalized, product representation: it gives an order containing o once the factor systems are taken integral.

A further question is to what extent the arithmetic division into domains can characterize the algebraic splitting fields. In other words: do different splitting fields always produce different domain decompositions, or already different principal domains? And do the principal domains of all splitting fields perhaps exhaust all maximal orders?

That such a characterization is certainly impossible by a single maximal order is shown by the simplest examples. For instance, the maximal order

$$\left[1, i, j, \frac{1+i+j+k}{2} \right]$$

of the quaternion field contains, besides the principal order $[1, i]$, also the principal order

$$\left[1, \frac{1 + i + j + k}{2} \right]$$

of the field of third roots of unity.

Göttingen, August 1932.